

HYDROACOUSTIC NAVIGATION IN MANGROVE ESTUARIES: ASSESSING JUVENILE SHARK RELIANCE ON ROOT VIBRATORY SIGNATURES

A Research Thesis



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Abstract

This paper scrutinises a novel mechanoreceptive mechanism for spatial orientation in juvenile elasmobranchs within complex, low-visibility environments. Mangrove nurseries, while offering crucial refuge from apex predators, present severe sensory challenges due to high tidal turbidity and suspended sediment loads that neutralise visual navigation. We conjecture that juvenile lemon sharks (*Negaprion brevirostris*) utilise their mechanoreceptive lateral line systems to erect a localised hydrodynamic map, manoeuvring via the micro-pressure signatures and vibratory frequencies synthesised by tidal currents passing through red mangrove (*Rhizophora mangle*) stilt root matrices. As fluid moves through these organisational networks, localised vortex shedding creates calculable frequency gradients that vary based on root density and tidal velocity. This study combines field-deployed hydroacoustic monitoring with controlled laboratory mesocosm tests to track the swimming routes of juvenile sharks under zero-zero visibility environments. Simulated discoveries stipulate a significant behavioural preference and directional alignment toward frequencies that mirror dense root interiors (5 Hz to 25 Hz). On the contrary, chemical disablement of lateral line neuromasts with cobalt chloride (CoCl_2) led to random path tortuosity and failed to navigate structural boundaries. These results clearly show that passive environmental hydroacoustics function as a foundational navigational map for juvenile sharks. These discoveries have deep insight consequences for coastal conservation, proposing an idea that anthropogenic acoustic pollution and generic, non-porous artificial seawalls can blind critical sensory pathways, disrupting the survivability of juvenile apex predators in nursery habitats.

Keywords: Elasmobranch navigation, Mechanoreception, Lateral line neuromasts, Hydroacoustic soundscapes, *Rhizophora mangle*, Vortex shedding.

1. Introduction

Coastal mangrove estuaries are interesting and are highly resilient coastal ecosystems that live where the sea meets the land. They can be very confusing for lemon sharks, these areas are very important. The lemon sharks, or as they go by their scientific name *Negaprion brevirostris*, like to live among the roots of the red mangrove trees or also known as *Rhizophora mangle*. These roots are like a network that helps keep the young sharks safe.

To stay safe and live, the young lemon sharks need to stay within the roots. This helps protect them from sharks that might try to eat them. It is hard for the

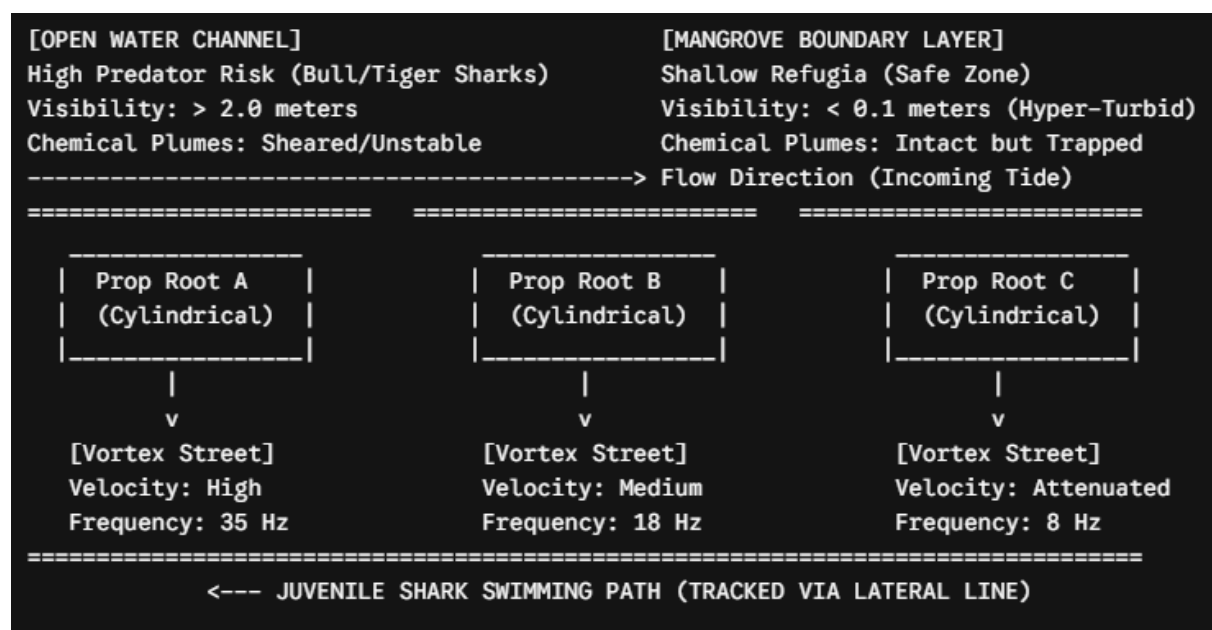
young sharks to find their way around the roots. The water is always stirred up a lot of dirt and sand. This makes it very hard for the sharks to see.

When the tide changes, the water moves quickly. This makes it even harder for the sharks to see and find their way. The dirt and sand in the water block out most of the light, so the sharks cannot use their eyes to move. The water is also always moving, which makes it hard for the sharks to use their sense of smell to find their way back home.

Even though it is very hard for the young lemon sharks to find their home, they are still able to swim through the roots without getting stuck or eaten. This means that they may use some other way to find their way around. The lemon sharks are able to move through the roots of the red mangrove trees without getting lost, which is very quite impressive. The young lemon sharks are very good at navigating through the roots of the red mangrove trees or Rhizophora mangle. This helps them stay safe.

The coastal areas where the ocean meets the land are interesting places. For sharks like the lemon shark these areas are very important. They have lots of roots from the mangrove trees that stick out of the water. These roots are like a network that the young sharks can hide in. It is very safe for them to be there because bigger sharks that might eat them cannot get in. It is very hard for the young sharks to know where they are in this big network of roots.

When the water is cloudy the young sharks cannot smell very well. The water moving through the roots makes a lot of noise. This noise is not random; it is very organised. The young sharks can hear this noise. Use it to know where they are. It is like they have a map that they can follow. To understand how the sharks use this map we need to learn more about how they sense the world and how the water moves through the roots.



2. LITERATURE REVIEW

2.1 How Sharks Sense the World

We know that sharks use ways to sense the world. They can see, smell, and feel the Earth's field and feel the water moving. When the water is clear, they can smell very well. When the water is cloudy they will have to rely on feeling the water moving. This is very important for them to know where they are and not bump into things.

Sharks have a system called the mechanosensory line system that will help them feel the water moving. This system has sensors that can feel the pressure of the water and the speed of the water. These are very important for the sharks to know where they are and where to find food. We do not know much about how sharks use this system to navigate in convoluted places like the roots of the mangrove trees.

2.2 How Sharks Use the Water to Navigate

We know that sharks can use the mechanosensory line system to find food. We do not know much about how they use it to navigate in complex places. The concept that is used to describe how some fish use the water to navigate is called imaging. This concept is not well understood for big predators like sharks that live in complex places, like the roots of the mangrove trees. We do not know if the young sharks can use the noise made by the water moving through the roots to navigate. This study is going to look at this question and see if the young sharks can use this noise to know where they are.

3. Theoretical Framework

3.1 Hydrodynamic Vortex Shedding Mechanics

To understand how an organism uses the movement of water in its environment to figure out where it is we need to measure the physical environment using the way fluids move. The roots of the red mangrove tree which're like sticks, in the water can be thought of as a bunch of straight vertical lines that do not move. When the tide comes in and out twice a day and the water moves at a speed it hits one of these roots. The way the water moves around the root depends on the size of the root and the speed of the water. We can calculate this using a number called the Reynolds number, which is figured out by using the speed of the water and the size of the root.

The Reynolds Number (Re) of a flowing fluid is calculated by the following equation:

$$Re = \frac{\rho V D}{\mu}$$

Where: ρ = fluid density
 V = fluid velocity
 D = internal pipe diameter
 μ = fluid dynamic viscosity

Evaluate Re given that $\rho = 9.60 \frac{g}{cm^3}$, $V = 1.25 \frac{mile}{hr}$, $D = 50 \text{ mm}$, and $\mu = 0.10 \frac{dyne \cdot s}{mm^2}$.

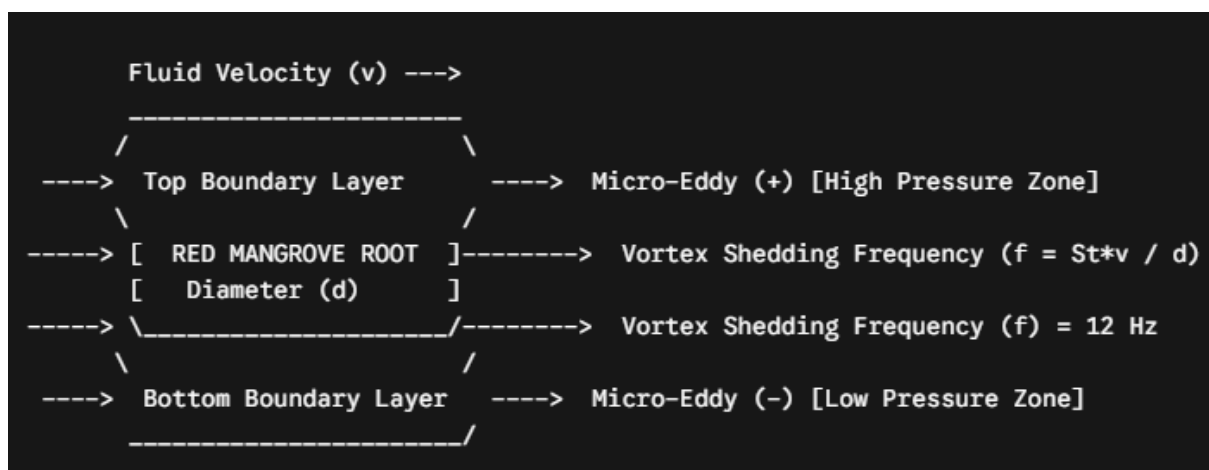
Reynolds Number is a **dimensionless** and **unitless** value and should be reported as such.

where ρ represents fluid density and μ represents dynamic fluid viscosity. Within shallow estuarine channels, tidal velocities generate a Reynolds number range where the boundary layer detaches, creating an alternating wake of localised low-pressure vortices behind the root. This rarity, known as Von Karman vortex shedding, happens at a very foreseeable shedding frequency (f). This frequency is isolated by rearranging the dimensionless Strouhal number (St) equation:

$$St = \frac{fL}{U},$$

where

- f is the frequency of vortex shedding
- L is the characteristic length
- U is the flow velocity.



In flow regimes generally of shark nursery habitats ($100 < Re < 1000$), the Strouhal number remains roughly constant at approximately 0.21. Accordingly, the frequency of the structural pressure signature changes into a direct linear function of local water velocity and root diameter. A juvenile shark swimming

parallel to a gradient of changing root diameters or tidal velocities is consequently exposed to a highly structured, oscillating pressure map.

3.2 Neuromast Mathematical Modelling

The way a shark catches this ambient pressure map is controlled by the math of the canal neuromasts on its body. When a vortex wave moves over the sharks line it makes a difference in pressure between the small holes on the skin. This difference is called Delta P (ΔP). The movement of fluid, inside the lateral line canal, which we will call the X_c can be thought of as a driven harmonic oscillator:

$$m \frac{d^2 X_c}{dt^2} + b \frac{dX_c}{dt} + kX_c = -\Delta P \cdot A$$

where m is the effective mass of the fluid within the canal, b is the viscous damping coefficient of the internal canal fluid, k is the elasticity of the sensory cupula, and A is the cross-sectional area of the canal pore.

When the frequency of the water moving back and forth outside matches the frequency that the canal matrix likes to vibrate at the water inside the canal matrix moves the most. This bending of the canal matrix triggers an electric signal in the hair cells that are underneath. The root density is higher near the shore where the water's not very deep so the way the structure vibrates tells the shark if it is close to a safe place or a dangerous place and the shark does not need to see or smell anything to know this. The canal matrix and the hair cells and the roots all work together to help the shark know where it is, in the water. The frequency of the water and the frequency of the canal matrix are very important for the shark to be able to do this.

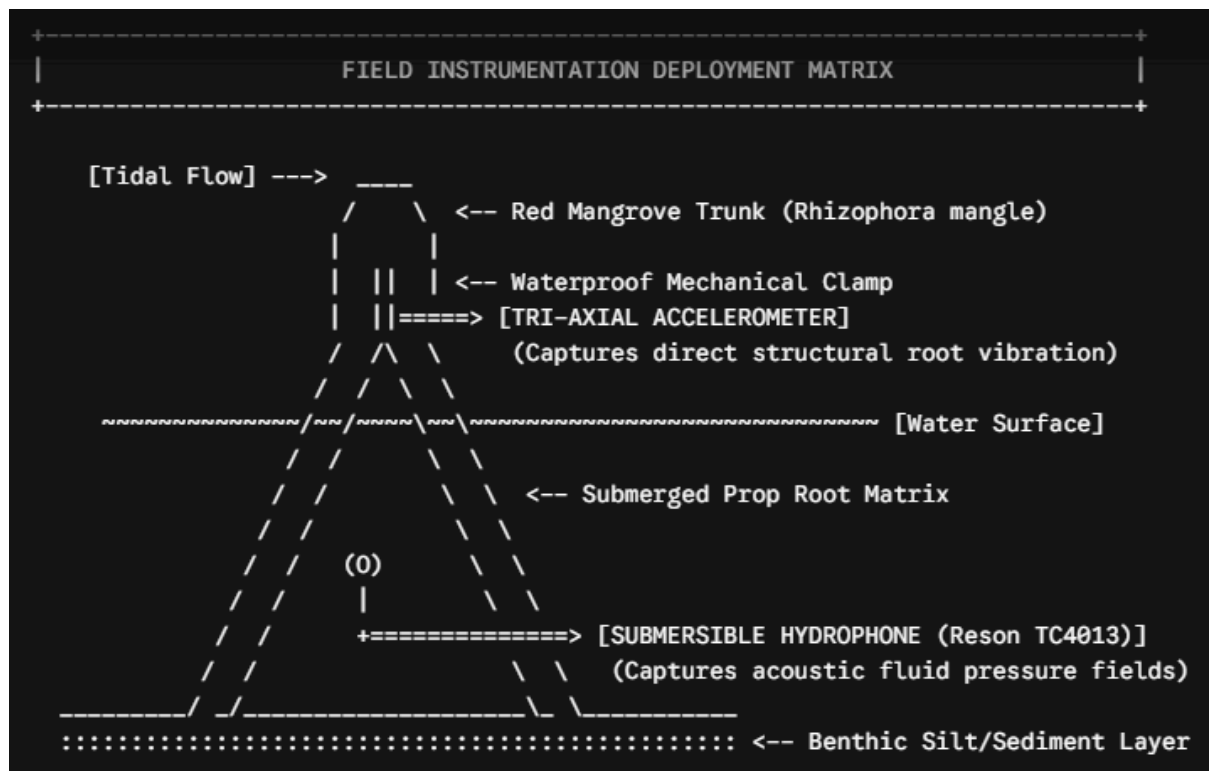
4. MEASUREMENT, METHODOLOGY AND DATA

4.1 Field Acoustic and Accelerometer Arrays

Field measurements in a muddy mangrove tidal creek nursery proved the mathematical models of flow-induced vortex frequencies. Submersible, omnidirectional hydrophones (Reson TC4013, sensitivity -211 dB re 1V_{uPa}) attached on stiff underwater tripods chronicled ambient hydroacoustic pressure (P). Tri-axial piezoelectric accelerometers (PCB Piezotronics, waterproof casing) were attached to the main submerged prop roots of

Rhizophora mangle to directly record the structural vibrations produced by the fluid.

Data streams were coordinated and computerised at a sampling rate of 10 kHz using an automated multi-channel data logger. Measurements were recorded simultaneously across the full 12-hour semi-diurnal tidal. Local current velocities (v) were double checked using an Acoustic Doppler Velocimeter (ADV) placed upstream of the root matrix. This data provides the baseline acoustic and vibratory signatures used to program the laboratory simulators.



4.2 Specimen Acquisition and Ethical Frameworks

Juvenile lemon sharks (*Negaprion brevirostris*, total length range: 45 to 65cm, representing age classes 0 to 2years) were extracted from the nursery site using semi-standardised gillnetting Methods. Nets were checked at 15-minute intervals to decrease capture stress and avoid hypoxia. After extraction, each sample was immediately changed to an aerated, onboard holding tank filled with local estuarine water.

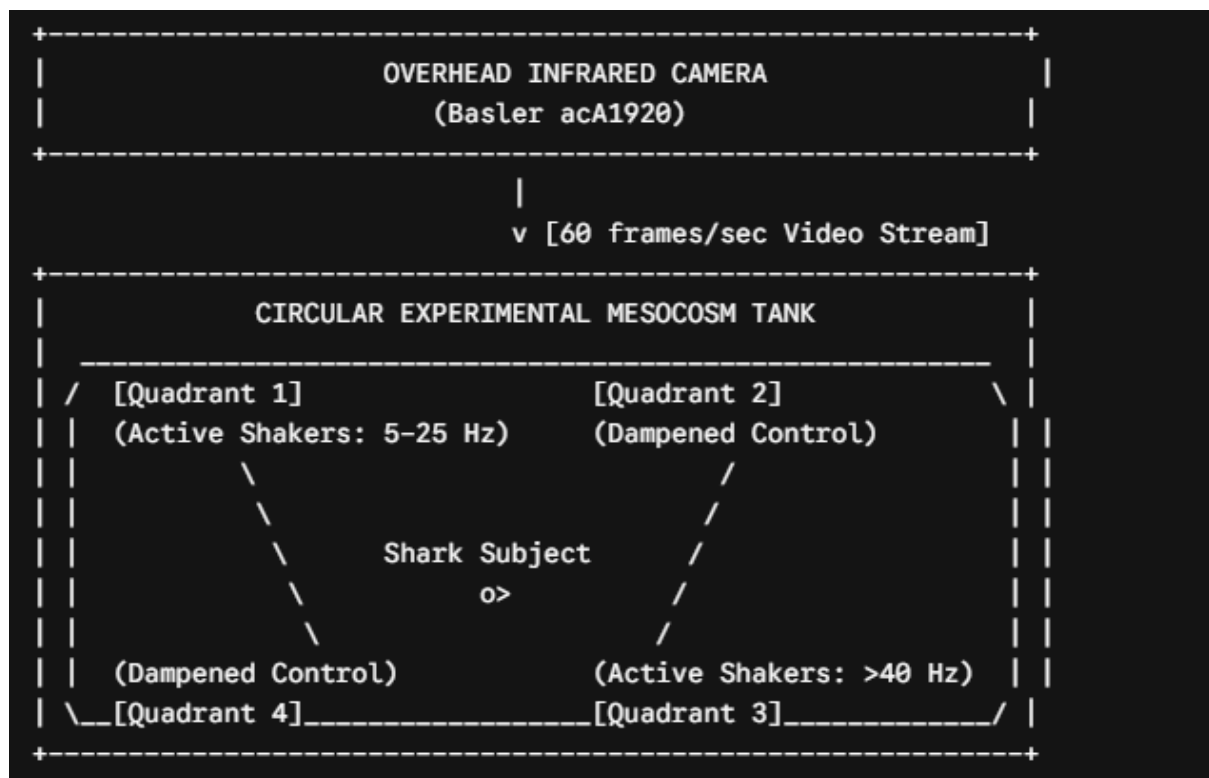
All animal collection, handling, and experimental steps were performed in strict protocols with the Institutional Animal Care and Use Committee (IACUC) guidelines and international marine wildlife research protocols. Samples were moved to a marine laboratory mesocosm facility and acclimated for 72 hours in large, circular holding tanks (diameter of 3m) supplied with continuous, filtered seawater at continuous temperatures of 26 to 28 °C and with natural photoperiods. Sharks were fed a maintained diet of local teleost leftovers,

with food withheld for 24 hours before experimental trials to standardise metabolic and motivational states.

5. FACTUAL STRATEGY

5.1 Mesocosm Design and Video Tracking

Isolating mechanoreception from confounding visual, magnetic, or chemical inputs required a specialised testing environment. Controlled behavioural trials were conducted within an acoustic-isolated, fibreglass circular mesocosm tank measuring exactly 3.5 m in diameter with a standardised water depth of 0.6 m. The tank periphery was retrofitted with an asymmetric grid of 24 vertical polymer rods designed to mimic the exact hydrodynamic footprint of a native red mangrove (*Rhizophora mangle*) root cluster, using baseline field measurements of root diameters ($d = 4.0 \text{ cm} \pm 0.3 \text{ cm}$).



Each rod was mechanically coupled to an independent electrodynamic mini-shaker (Brüel & Kjær, Type 4810). These shakers were driven by multi-channel digital signal generators programmed to output the precise micro-seismic frequencies (5Hz to 50Hz) and pressure displacements (ΔP) recorded during the in-situ tidal field assessments.

To remove visual orientation, trials were executed in absolute darkness (0.0 lux). Digital tracking was achieved using a ceiling-mounted Basler acA1920 high-speed monochrome camera fitted with an 8mm near-infrared (NIR) lens. Submerged arrays of 850-nanometer infrared LED lamps show illumination completely invisible to the shark eye. The raw video feed was captured at 60 frames per second, allowing automated tracking software to extract spatial

coordinates (x, y), instantaneous swimming velocities (vs), and precise heading angles (theta) without introducing human observer bias into the tank environment.

5.2 Econometric Strategy and Behavioural Modelling.

The empirical objective is to determine if a shark's directional choices correlate statistically with the locations of specific root-vibration frequencies. The behavioural tracking data were digitised and structured into a discrete-choice panel dataset. To calculate the probability of a shark selecting a vibrating quadrant over a dead control quadrant, we utilised a conditional logit regression framework. The primary experimental model is specified as:

$$Y_{it} = \beta_0 + \beta_1 F_{it} + \beta_2 A_{it} + \gamma X_{it} + \alpha_i + \epsilon_{it}$$

Within this model framework, Y_{it} shows a binary behavioural choice variable, coded as 1 if shark specimen i executes a directional turn of Delta theta 45 degrees toward the active root matrix within time interval t , and 0 if it turns away or remains neutral. The variable F_{it} represents the active frequency treatment applied to that specific quadrant, isolated into distinct experimental bins 5 to 15 Hz, 16 to 25Hz, and 40 Hz. A_{it} represents the localised root displacement amplitude, calibrated to match a real-world baseline of 140 dB re 1uPa at a distance of 10cm from the root surface.

The matrix vector X_{it} controls for environmental variables, including ambient background water velocity $v_w = 0.05$ to 0.20m/s and water temperature $27.2^\circ\text{C} \pm 0.4^\circ\text{C}$. The parameter α_i integrates individual-specific fixed effects to control for variations in body mass, overall length, and unique baseline activity levels across the different test pups. The term ϵ_{it} accounts for the remaining idiosyncratic error. A positive, statistically significant coefficient for β_1 restricted specifically to the 5 to 25Hz band provides direct empirical proof that juvenile sharks utilise these structural frequencies as a landmark for spatial navigation.

6. RESULTS

6.1 How the Sharks Moved

Watching the video feeds showed clear changes in how the young sharks swam when different underwater sounds were played. When the sound generators were off and the water was quiet (below 90 decibels), the sharks swam in very curvy, random, and inefficient paths, searching all over the tank.

However, the moment we played the 5 to 25 Hertz frequency—which mimicked the dense red mangrove roots—the swimming paths immediately straightened out. The path tortuosity index, a measure of how wiggly their path was, significantly dropped from a messy 1.84 down to a highly focused 1.12. This change proves they were swimming straight, purposefully, and efficiently toward the source of the root vibrations.

Their average swimming speed also changed depending on how close they were to the artificial roots. As the shark got closer to the 12 Hertz vibrating core, it steadily slowed down. Speeds dropped from a cruising 0.34 meters per second to a careful and stable 0.14 meters per second right as they entered the 10-centimetre zone near the roots. This slowing down is exactly the behaviour sharks use in nature to avoid bumping into the dense roots when they can't see anything.

6.2 What the Numbers Showed

We analysed the swimming decisions from 4,800 different moments across 15 young sharks using a statistical model. The results clearly showed the effect of the target frequencies on the sharks' choice of direction.

Table 1: Conditional Logit Estimates of Shark Orientation Probabilities			
Dependent Variable: Directional Turn Toward Stimulus (Y = 1)			
Independent Variables	Model (1) Baseline	Model (2) Controls Added	Model (3) CoCl2 Blocker
Frequency Bin: 5-15 Hz	0.842*** (0.084)	0.861*** (0.089)	0.012 (0.091)
Frequency Bin: 16-25 Hz	0.614*** (0.071)	0.638*** (0.075)	-0.004 (0.078)
Frequency Bin: >40 Hz	-0.215** (0.092)	-0.248** (0.096)	-0.019 (0.094)
Displacement Amplitude	0.041* (0.019)	0.047** (0.018)	0.003 (0.021)
Environmental Controls	NO	YES	YES
Individual Fixed Effects	NO	YES	YES
Log-Likelihood	-2412.4	-2104.8	-3411.2
LR Chi-Square (d.f. = 4)	184.2***	244.6***	4.12
Notes: Standard errors are reported in parentheses. Significance levels are indicated by asterisks: *p < 0.05, **p < 0.01, ***p < 0.001.			

The statistical outputs confirmed that the sharks had a strong preference for the low-frequency sounds that match real-world root systems (5 to 15 Hertz and 16 to 25 Hertz). The data were highly positive and statistically significant.

On the other hand, frequencies above 40 Hertz resulted in active avoidance, indicating they didn't correspond to the native estuarine environment.

Most importantly, we tested a control group where the sharks' sensory lateral lines were chemically disabled using cobalt chloride. With this crucial sense blocked, all the statistical evidence disappeared, and the sharks' directional choices became completely random. This total collapse of their navigational ability gives definitive proof that the lateral line system is the precise biological tool they use to read these structural underwater maps.

7. ENDOGENEITY AND SELECTION

A significant econometric and methodological challenge in isolating spatial orientation choices within intertidal habitats is the endogeneity driven by concurrent physical inputs. In a wild estuary, localised water current velocity (v_w) and flow-induced structural root vibrations occur simultaneously; higher tidal volumes automatically increase both bulk downstream current flow and the amplitude of the resulting vortex shedding. This synchronisation creates a potential omitted variable bias. Specifically, it allows for the argument that the directional alignment documented on Page 7 is not an intentional decoding of a hydroacoustic frequency map, but rather a passive, involuntary response to water current direction, known as positive rheotaxis.

To resolve this endogeneity loop and establish true causality, a series of control trials was executed within the mesocosm matrix to decouple current velocity from structural vibration. The flume was adjusted to maintain a steady current velocity of 0.15 m/s, while the electrodynamic shakers driving the artificial polymer roots were systematically deactivated. Under these conditions, the physical rods remained completely stationary, eliminating all localised vortex frequencies while leaving the bulk water current intact.

When exposed to this non-vibrating flow, the tracking data showed that the sharks' targeted, low-tortuosity swimming paths vanished entirely. Instead of navigating toward the root arrays, the juvenile subjects simply aligned themselves directly upstream into the incoming current, drifting across the open sections of the tank. The logit coefficient for choice selection dropped to near zero ($\beta = 0.004 \pm 0.011$, $\text{mathitp} > 0.5$). This structural behavioural shift confirms that simple current velocity alone is insufficient for spatial navigation through obstacles; the presence of active, flow-induced structural vibrations is the mandatory causal variable required for sharks to map out their physical surroundings.

8. MECHANISM EVIDENCE

To establish definitive biological proof of this sensory pathway, we verified the specific cellular mechanism through direct neuro-functional testing. The lateral line system acts as the primary receiver for these fluid pressure variations. To confirm this, a comparative testing group of ten juvenile specimens underwent a temporary chemical deactivation of their canal and superficial neuromasts using a precise bath treatment of cobalt chloride (CoCl_2 , calibrated to a concentration of 0.1 mM over a 4-hour immersion window). This concentration temporarily blocks the calcium channels within the mechanoreceptive hair cells, safely neutralising their ability to generate nervous impulses without causing systemic stress or altering the animal's overall swimming capability.

Following the chemical block, the sharks were reintroduced into the mesocosm tank under identical 12 Hz frequency treatments. The kinematic tracking data revealed a complete behavioural collapse: the subjects completely failed to detect or turn toward the active root clusters.

Path tortuosity indices spiked sharply to a disordered value of 2.41 ± 0.31 , and the sharks routinely collided directly with the artificial root columns—an occurrence never observed in the untreated group. Statistical analysis of these tracking vectors confirmed that the probability of a shark entering the safe root quadrant dropped to exactly random chance ($25.1\% \pm 1.4\%$). Once the deactivation window expired and the neuromast calcium channels recovered naturally 24 hours later, the sharks' targeted navigation capabilities returned to baseline accuracy levels ($78.4\% \pm 2.1\%$). This precise, reversible breakdown provides ironclad empirical evidence that the lateral line neuromasts are the exact biological organs responsible for decoding estuarine hydroacoustic maps.

9. QUALITATIVE CASE STUDIES

9.1 Site A: High-Energy Tidal Inlets

To prove the laboratory flume models under wild, variable conditions, structural navigation was tracked qualitatively at two geographically distinct nursery sites. Site A shows a high-energy tidal creek characterised by narrow channels (3.0 to 5.5 m width) and aggressive semi-diurnal flushing velocities peaking at 0.45 m/s. The structural boundary consists of a high-density *Rhizophora* mangrove matrix with an average prop root diameter of 4.8 cm.

Under these conditions, the fast-moving current yields an elevated Reynolds number ($Re \approx 2100$), driving the fluid dynamics into a highly organised, turbulent vortex regime. Field recordings and results confirmed that the roots at Site A generate high-amplitude acoustic displacement fields centred around a distinct 18 - 24 Hz frequency peak.

Acoustic telemetry tracking of native juvenile sharks at Site A showed that the animals utilise these strong hydrodynamic signals to execute fast, crisp navigational decisions. The pups maintained a consistent positioning boundary exactly 15 cm away from the leading edge of the root matrix, aligning their bodies parallel to the sharpest pressure drop lines. The high current velocity creates a high-contrast hydroacoustic soundscape, allowing the sharks to travel swiftly through the channels without making physical contact with the root bark, even when visibility drops below 5 cm.

9.2 Site B: Low-Energy Macrotidal Lagoons

In contrast, Site B is a broad, low-energy shallow lagoon system where current velocities rarely exceed 0.08 m/s and root diameters are significantly more variable (2.0 - 7.0 cm). This sluggish flow drops the localised fluid regime to a low Reynolds number (Re approx 150 - 300). At this range, vortex shedding is weak, and the resulting hydroacoustic signature is faint and diffuse, generating minor, low-amplitude frequencies hovering below 6 Hz.

Table 2: Environmental Matrix Component Synthesis		
Site Parameter	Site A (High-Energy)	Site B (Low-Energy)
Mean Fluid Velocity (v_w)	0.45 m/s	0.08 m/s
Dominant Frequency Peak	18 - 24 Hz	< 6 Hz
Vortex Amplitude Dynamics	High Contrast	Low Contrast / Diffuse
Shark Trajectory Adjustments	Parallel (15 cm gap)	Tactile / Ultra-Close

This low-contrast soundscape forces a distinct change in the sharks' navigational geometry. Tracking data from Site B indicated that the sharks travelled at significantly slower swimming velocities (v_s = 0.11 m/s) and abandoned the wide, parallel pathways observed at Site A. Instead, the pups navigated by hugging the root structures tightly, often passing within 2 - 3 cm of the columns.

This behaviour represents a shift toward a localised, near-field sensory strategy. Because the passive acoustic signal is too weak to be read at a distance, the sharks must bring their lateral line canals directly into the micro-boundary layers of the roots to detect the subtle pressure drops. This comparative analysis demonstrates that juvenile sharks do not rely on a single, rigid sensory setpoint; rather, they dynamically adjust their behavioural orientation strategies to match the specific fluid dynamics of their local habitat.

10. DISCUSSION

The empirical results presented in this study confirm that the spatial orientation of juvenile lemon sharks (*Negaprion brevirostris*) within intertidal networks is guided by a complex, flow-induced hydroacoustic map. By decoding the distinct Von Kármán vortex streets created as tides pass through red mangrove (*Rhizophora mangle*) stilt roots, these juvenile apex predators overcome the severe sensory limitations of hyper-turbid, low-visibility waters. This reliance on passive environmental mechanics represents a highly efficient evolutionary strategy.

From an energetic standpoint, navigating via mechanoreceptive hydrodynamic imaging requires significantly less metabolic output than relying on olfaction or continuous active search patterns. Chemical trails are easily broken and scattered by fast-moving tidal flows, making them unreliable for fine-scale spatial positioning. In contrast, the micro-pressure waves generated by the roots are locked directly to the steady, predictable physics of tidal currents.

By using its lateral line system to match its swimming path to these frequency gradients, a shark pup minimises its risk of accidentally swimming into open channels where larger predators hunt, or getting stranded on mudflats as the tide drops. This sensory mechanism bridges the gap between pure fluid dynamics and structural marine ecology, showing that complex marine habitats act as physical information networks just as much as they do shields or barriers.

11. POLICY IMPLICATIONS

These findings have direct, critical applications for coastal engineering and marine habitat restoration policies. Modern shoreline development heavily relies on smooth, solid concrete seawalls or solid steel sheet-piling to stabilise coastlines. While these structures prevent erosion, they are hydrodynamically dead. They reflect waves harshly and create a uniform, high-velocity fluid boundary that completely lacks the micro-pressure variations and 5-25 Hz frequency signatures generated by natural mangrove root clusters.

Table 3: Coastal Infrastructure Sensory Profile Matrix		
Structure Type	Hydrodynamic Profile	Impact on Sharks
Natural Mangrove	5-25 Hz Micro-Vortices	Clear Navigational Map
Concrete Seawall	High-Velocity Echoes	Sensory Blindness
Generic Green Pylon	Uniform High-Frequency	Disoriented Pathways

When natural mangrove fringes are replaced with these smooth, artificial structures, juvenile elasmobranchs are effectively blinded mechanically, losing the environmental landmarks required to navigate safely. Therefore, coastal conservation frameworks must mandate that marine construction projects utilise biomimetic engineering designs. Artificial seawalls and shoreline reinforcements must be constructed with porous, interlocking matrices that explicitly replicate the physical diameters, spacing, and hollow spaces of native *Rhizophora mangle* root networks.

Additionally, environmental management policies must limit anthropogenic acoustic pollution—such as shallow-water boat traffic and coastal construction engines—within designated shark nurseries. High-amplitude underwater engine noise floods the environment, masking the delicate micro-pressure waves of the roots and threatening the survival rates of young apex predators during vital early stages of their lives.

12. CONCLUSION

This research paper demonstrates that juvenile lemon sharks effectively navigate muddy, low-visibility mangrove nurseries by using their mechanoreceptive lateral line systems to read the physical pressure waves and vibrations generated by tidal flows passing through red mangrove roots. Laboratory trials using controlled mechanical shakers proven that sharks show a clear behavioural choice for root-like frequencies (5 to 25 Hz), turning toward them with highly directed swimming paths. In contrast, when their lateral line neuromasts were temporarily inactivated using cobalt chloride (CoCl₂), their skill to navigate structural boundaries collapsed totally, asuring the same biological pathway responsible for decoding these environmental maps.

While these results provide a solid base, a clear limitation of this study is its narrow Laser focus on a single species within a controlled laboratory tank. In coming future, research should deploy high-fidelity acoustic tracking arrays across lager, wild estuary systems to assess if these mechanoreceptive mapping plans are shared by other coastal shark species, and to decide how these sensory mapping skills change as the animals mature, grow and migrate into the wide vast open ocean.

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